Working with Image Data

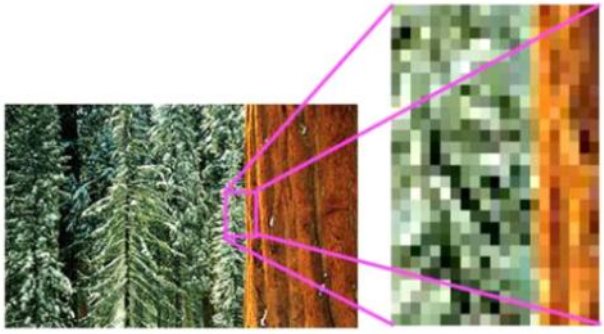
* Image :-

It is a visual representation of an object.

Digital Image :-

↳ It is a visual representation that is made up of pixels.

Pixels :- Tiny blocks that creates a digital image.



→ These tiny blocks are pixels.

While looking at the image, these tiny blocks are pixels but in reality pixels are nos.

* Representation of Image :-

↳ Images are represented using pixel values

↳ It is a matrix of pixels.

↳ A pixel matrix can be represented using an array.

↳ Each pixel has a specific colour and when these pixels forms a matrix, they create an image.

* Colour Spaces :-

↳ A way to represent colours in a digital image.

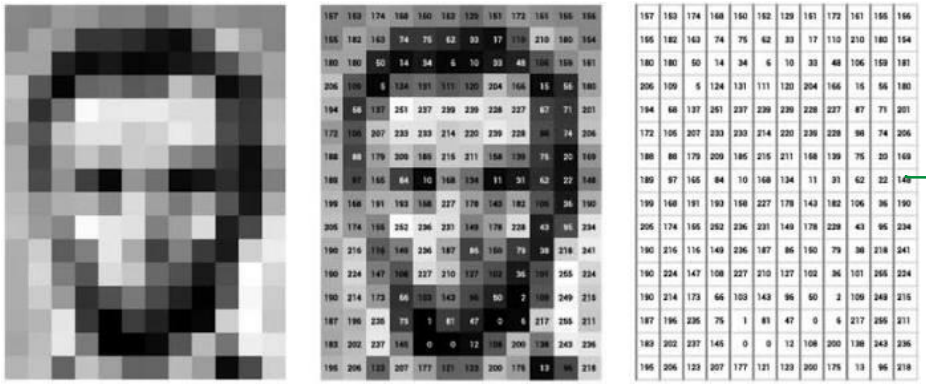
↳ A way to represent colours in a digital image.

Gray Scale (Black & White)

RGB (Red, Green, Blue)

Gray Scale :- Each pixel is represented by a single no. that shows how dark or light the pixel is.

This value ranges from 0 (black) to 255 (white)



pixel matrix or array

38 ⇝ dark shade

237 ⇝ light shade

Colour Image :- Each pixel is made up of 3 colour [R, G, B]

Each colour is represented by a no.

Red → 0 to 255
Green → 0 to 255
Blue → 0 to 255

(R, G, B) → 3 nos. are one over the other

↓

These 3 nos. combined to form a final pixel of coloured

pixel of coloured image

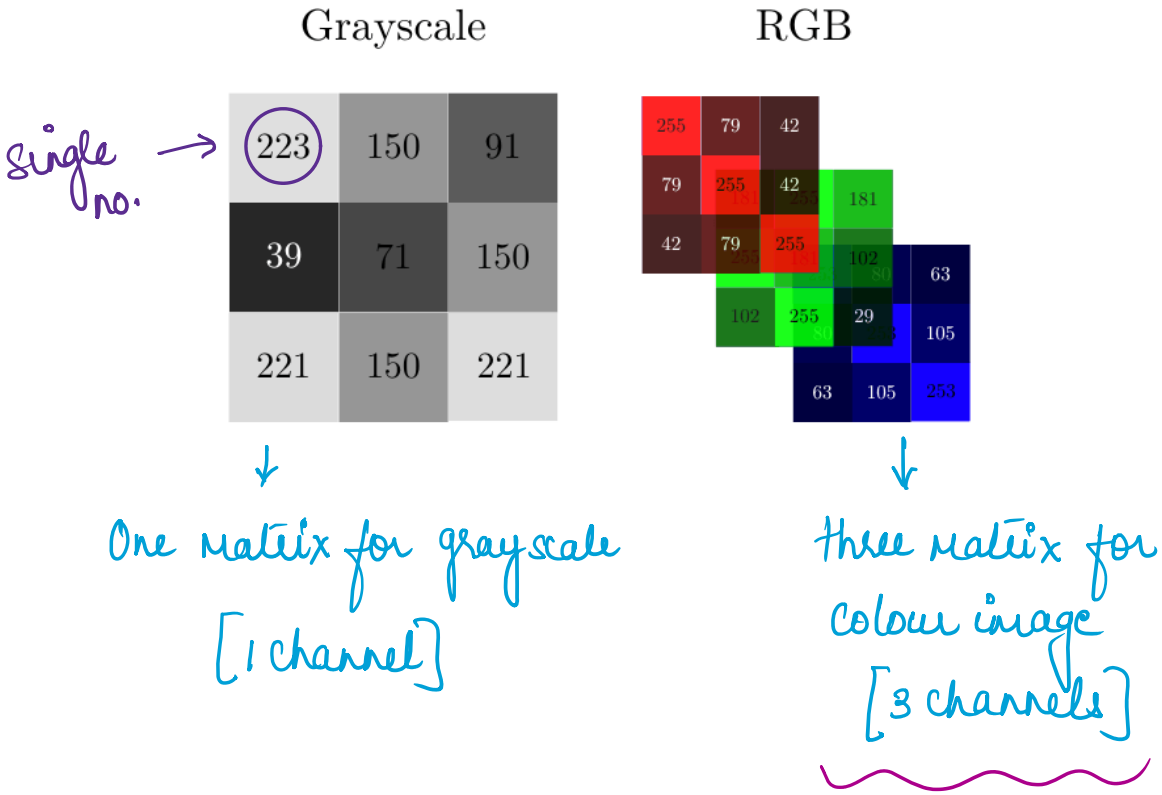
Ex- for pure red → R G B
255 0 0

for pure blue → R G B
0 0 255

for pure green → R G B
0 255 0

Some values of red, green & blue constitutes a coloured image.

Ex- Coloured Img pixel → (105, 94, 201)



One matrix for grayscale [1 channel]

three matrix for colour image [3 channels]

upon mixing these values of the 3 channels, the coloured image is formed.

Ex-



- A grayscale image measures the intensity (brightness) of a pixel that is why we use only 1 matrix
- A standard colour image needs 3 matrices (R, G, B) to represent 1 image.

* Why values from 0 to 255 are taken ?

In computer machines, everything is stored in the form of bits

1 bit → 0 or 1

1 byte → 8 bits

The pixel values are stored in →

2^8 bits = 256 ⇝ 0 to 255